

HYVINKAA HYVINKAANKYLA, Finland

WMO: 028290

Lat: 60.5958N

Lon: 24.8031E

Elev: 287

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.5	(c) -3.3	(d) -13.9	(e) 2.6	(f) -9.0	(g) -7.6	(h) 3.7	(i) -2.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	81.8	66.2	78.3	63.9	74.9	62.4	68.9	77.7	66.6	75.1	64.3	72.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	96.2	73.1	63.3	88.0	70.4	61.1	81.2	68.0	33.2	78.0	31.3	75.0	29.6	72.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -13.5	(f) 84.4	(g) 9.3	(h) 4.6	(i) -20.2	(j) 87.7	(k) -25.6	(l) 90.4	(m) -30.9	(n) 92.9	(o) -37.6	(p) 96.2		
(4) N/A	(5) N/A	(6) N/A	(e) -13.8	(f) 70.6	(g) 9.3	(h) 2.9	(i) -20.5	(j) 72.7	(k) -25.9	(l) 74.4	(m) -31.1	(n) 76.1	(o) -37.8	(p) 78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	19.8	23.1	28.6	39.3	50.8	57.5	63.0	59.7	51.9	40.3	34.2	27.2		
	DBStd	16.41	11.64	11.14	8.24	6.17	7.51	6.25	5.74	5.09	5.83	6.48	7.65	9.76		
	HDD50	4309	937	755	664	326	81	11	0	1	47	305	474	708		
	HDD65	8700	1402	1175	1129	771	443	236	110	178	394	765	924	1173		
	CDD50	1157	0	0	0	4	105	236	402	302	103	5	0	0		
	CDD65	74	0	0	0	0	2	12	47	13	0	0	0	0		
	CDH74	885	0	0	0	0	65	178	497	145	0	0	0	0		
	CDH80	163	0	0	0	0	6	23	106	28	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	24.5	2.0	1.5	1.3	1.3	1.5	2.5	2.8	2.8	2.2	2.5	2.3	1.9		
(15) Precipitation	PrecMax	31.5	3.6	3.7	2.7	2.8	3.1	4.6	8.9	5.7	5.3	6.3	6.4	4.6		
	PrecMin	18.8	0.3	0.0	0.2	0.2	0.1	0.6	0.0	0.4	0.6	0.6	0.2	0.2		
	PrecStd	3.3	0.8	0.9	0.6	0.7	0.8	1.1	1.8	1.2	1.3	1.3	1.3	0.9		
	PrecStd	3.3	0.8	0.9	0.6	0.7	0.8	1.1	1.8	1.2	1.3	1.3	1.3	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	42.4	50.8	67.8	80.8	83.2	87.0	83.8	71.6	57.3	50.5	47.0		
		MCWB	37.9	37.2	41.3	51.6	60.3	65.8	68.9	66.5	61.3	53.0	48.7	45.5		
	2%	DB	37.3	38.5	45.1	62.4	75.4	78.8	83.2	77.7	67.4	54.3	48.1	42.9		
		MCWB	36.3	36.3	38.7	47.9	58.9	62.4	67.9	65.0	59.5	51.4	47.3	41.5		
	5%	DB	35.5	36.4	41.5	57.1	71.2	75.4	79.9	74.0	64.3	52.5	45.5	40.0		
		MCWB	35.0	35.2	36.8	46.2	56.5	61.2	66.3	62.7	57.2	50.3	44.6	39.0		
	10%	DB	33.5	34.8	38.4	52.2	66.9	71.4	76.2	70.7	61.8	50.7	43.7	37.7		
		MCWB	32.8	34.1	35.0	42.9	53.9	59.1	64.5	61.0	56.9	48.7	42.9	36.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	38.4	42.3	52.9	63.9	68.3	72.0	71.3	62.9	54.8	49.2	45.9		
		MCDB	38.8	40.4	47.1	65.0	74.1	78.4	81.9	79.4	68.7	56.2	50.0	46.6		
	2%	WB	36.7	36.7	39.6	49.7	60.8	65.5	69.9	66.9	60.8	52.6	47.2	41.7		
		MCDB	37.3	38.1	43.6	59.7	69.7	76.2	79.3	74.0	65.1	53.7	48.1	42.8		
	5%	WB	35.0	35.4	37.5	46.7	58.2	62.5	68.1	64.5	59.3	50.9	44.8	39.0		
		MCDB	35.6	36.3	40.7	55.0	68.5	71.8	77.1	71.0	62.9	52.0	45.4	39.8		
	10%	WB	33.1	34.1	35.7	44.2	55.3	60.2	65.9	62.6	57.5	48.8	43.0	37.0		
		MCDB	33.5	34.7	37.8	50.5	64.9	69.2	74.1	68.3	60.5	50.2	43.6	37.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.2	14.2	17.9	21.1	18.9	18.5	18.2	15.0	11.6	7.1	7.7		
		MCDBR	6.7	8.7	16.8	28.3	29.0	26.1	24.0	24.3	19.5	14.4	8.4	8.7		
		MCWBR	6.6	7.4	11.2	15.8	14.2	12.4	11.0	12.6	12.4	11.4	8.1	8.2		
	5% WB	MCDBR	6.6	7.2	14.4	24.9	24.7	23.4	21.2	20.0	16.8	11.2	8.3	8.2		
		MCWBR	6.7	6.3	10.4	14.8	13.0	12.1	10.7	11.4	11.3	10.1	8.2	7.9		
			(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.304	0.329	0.358	0.355	0.352	0.374	0.365	0.339	0.327	0.294	0.379		
	taud		2.122	2.185	2.256	2.368	2.428	2.461	2.419	2.441	2.509	2.403	2.209	2.622		
	Ebn at Noon		149	213	250	266	276	279	268	262	250	208	138	103		
	Edh at Noon		13	23	29	32	32	32	32	29	23	18	12	9		
(44) All-Sky Solar Radiation	RadAvg		77	258	728	1211	1632	1696	1643	1265	769	336	94	45		
	RadStd		14	55	94	124	177	137	154	135	102	56	18	7		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	-254	N/A	N/A	N/A	N/A

Nomenclature: See separate page